

**Scalable Demographic Bias Auditing in Data Preprocessing Pipeline
using a Hybrid WOA-DE Algorithm for Optimized Metadata Provenance**

A Thesis

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CHAPTER 1

THE PROBLEM AND ITS SETTING

This chapter contains the introduction, the theoretical and conceptual framework, statement of the problem, hypothesis, the significance of this study, scope and limitation, and operational definitions.

Introduction

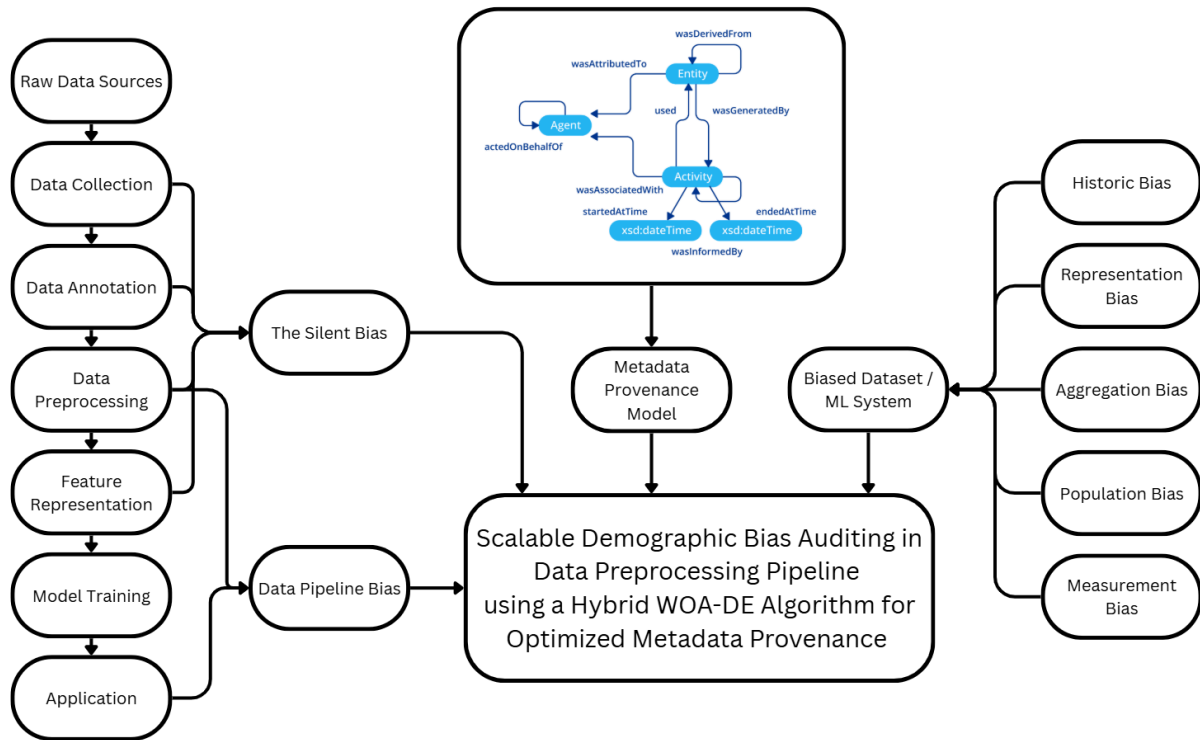
Introduce background about the relevance of bias

Introduce metadata provenance

Introduce metaheuristic algo (woa-de, woa, de)

Theoretical Framework

Figure 1.1 Theoretical Framework



Data Pipeline Bias and the Silent Bias

Kumar-B et al. (2021) investigated how bias enters and propagates through NLP and ML data pipelines. Their study reveals that the data pipeline itself is a primary source of bias or unfairness. It identified multiple stages where bias can occur such as: data collection, in which sources may overrepresent or underrepresent groups. Data annotation, in which human annotators bring their own subjective opinions or biases. Data preprocessing or cleaning, in which certain groups are unintentionally excluded. Feature representation, in which embeddings can encode historical or social biases. Model training and evaluation, in which bias from previous pipeline propagates to

prediction. Lastly, Applications, in which biased outcomes become visible to the users. This supports the researcher's claim that bias is a persistent issue throughout the data pipeline

An article by Trivedi, J. (2025) corroborates the "Data Pipeline Bias" as he highlighted that bias enters earlier in the data pipeline silently. This includes the early stages of: data collection, data annotation or labelling, data cleaning, and feature representation. The bias is considered "silent" because it is rarely recognized at the time it is introduced in the pipeline. Furthermore, when the silent bias propagates further downstream in the pipeline, it is harder to detect and correct later in the process. Both studies by Kumar-B and Trivedi confirm that biases do really exist from the beginning to the end of the data pipeline.

Metadata Provenance Model

Metadata Provenance shows the origin, history, and transformations of data. The article by Khan, U. H. (2025) discusses how data provenance works. First, the metadata systems record data history. Second, the blockchain or distributed ledger makes records tamper-resistant. Third, cryptographic hashing is used to detect changes. Fourth, AI/ML tools are used to automate provenance capture. Lastly, cloud-based solutions are used for centralized tracking. The diagram shows how data flows through transformation and who and what was involved. This tracks where the data came from, what processes are used to change it, who was responsible for the process, and when did it happen. This study is relevant as it proves why a data provenance is useful and trustworthy in detecting biases.

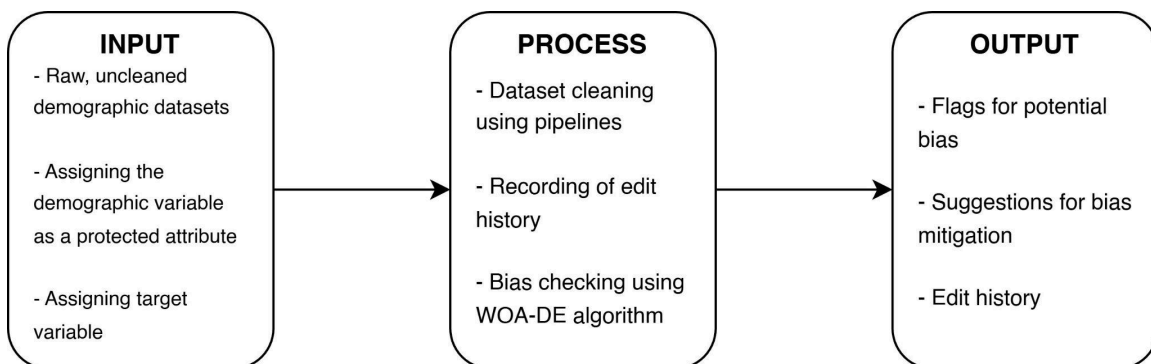
Biased Dataset / Machine Learning System

The study by Yang et al. (2024) investigates the causes of biases in AI models in the data pipeline from data collection to model development. They have pinpointed various sources of biases in AI systems that can lead to biased datasets and machine learning systems. This includes historic bias, which exists because of past societal prejudices or inequalities. Representation bias, which happens when some groups are underrepresented or overrepresented in the dataset. Aggregation bias, which exists when data from different groups are combined even though there's a difference between them. Population bias, which exists when the population used to train the model is not the same on where it is applied. Lastly, measurement bias, which occurs when the tools or methods used to collect data are inaccurate. This study is relevant as it proves that there are different kinds of bias found in the data pipeline that needs to be addressed.

Conceptual Framework

The conceptual framework of this study provides a structured approach for detecting demographic bias within data pipelines. The system operates by transforming raw, unverified data into audited, traceable datasets through a hybrid algorithmic approach.

Figure 1.2: Conceptual Framework



1. Input

The first phase consists of the raw materials and parameters required for the audit.

- **Raw, Uncleaned Demographic Datasets:** The system ingests tabular data (e.g., CSV files) containing missing values, inconsistent formatting, or latent biases. These represent the real-world "dirty" data typically found in initial pipeline stages.
- **Assigning Protected Attributes:** The user identifies specific demographic variables (e.g., Age, Gender, Race) that must be monitored for ethical violations.
- **Assigning Target Variable:** The researcher defines the outcome variable (e.g., Loan Approval or Hiring Decision). This is critical for calculating the mathematical success rates required for bias auditing.

2. Process

This phase represents the core technical innovation where the “Active Provenance” and “Hybrid Optimization” occur.

- **Dataset Cleaning using Pipelines:** As data undergoes transformations (e.g., handling null values or standardizing strings), the system monitors these changes in real-time.
- **Recording of Edit History (Metadata Provenance):** Instead of just saving the final file, the system captures Optimized Metadata. This creates a chronological "paper trail" or lineage of every edit, recording who made the change, what was changed, and when.
- **Bias Checking using WOA-DE Algorithm:** This is the analytical engine. The Whale Optimization Algorithm (WOA) performs a global search to scout for

high-bias clusters within the metadata logs, while Differential Evolution (DE) pinpoints the exact transformation that caused the ethical shift. This hybrid approach ensures the search is scalable and efficient even with large-scale data.

3. Output

The last phase provides the final audit results and evidence of the system performance.

- **Flags for Potential Bias:** The system generates automated alerts when the Statistical Parity or Disparate Impact thresholds are breached.
- **Suggestions for Bias Mitigation:** Based on the detected disparity, the system provides actionable feedback to the developers to correct the pipeline logic.
- **Edit History (Validated Traceability Logs):** The final deliverable is a comprehensive provenance report. This serves as a "Security Audit Trail," allowing stakeholders to trace a biased result back to its specific origin in the pipeline, ensuring accountability and transparency.

Statement of the Problem

The intent of the study is to identify and detect bias using metadata provenance and WOA-DE Algorithm to determine the exact cause of bias. Specifically, this study aims to answer the following questions:

1. What is the performance of the hybrid WOA-DE algorithm in bias auditing compared to standalone WOA and standalone DE in terms of:
 - 1.1. Accuracy
 - 1.2. Scalability
2. What traceability does a metadata provenance system provide in identifying the specific transformation that introduced a bias?

Hypothesis

Scope and Limitation of the Study

Significance of the Study

The purpose of the study is to provide a new way to handle fairness in automated systems that use demographic data. This study will also benefit:

Data Engineers and Developers. Since the system records an edit history or metadata provenance, it creates a chronological paper trail of every change made during the data cleaning process. This saves time because a biased result can be traced back to its specific origin in the pipeline, which makes it much easier to fix the logic compared to checking everything manually.

AI Ethics Auditors and Organizations. By using the hybrid WOA-DE algorithm, the audit process becomes more efficient and scalable even when dealing with large datasets. This

ensures that organizations can maintain transparency and accountability by having a validated traceability log for their data.

General Public. The system helps prevent ethical violations by monitoring specific demographic variables like age, gender, and race to ensure people are not treated unfairly.

Future Researchers. This study serves as a guide for using metaheuristic optimization and metadata to solve computing problems related to AI ethics. It fills a gap in current research by focusing on the active provenance of data instead of just looking at the final output.

Definition of Terms

The following terms have been used operationally in this study:

WOA (Whale Optimization Algorithm). An algorithm that acts as the “Global Explorer”. It mimics the spiral bubble hunting behavior of humpback whales. It searches through the massive metadata for potential clusters of bias.

DE (Differential Evolution). An algorithm that acts as the “Local Exploiter”. It uses “mutation” and “crossover” to fine-tune the search. It pinpoints the exact combination of features causing the bias.

AI Ethics. Involves detecting bias in training data to ensure AI systems are transparent, fair, accountable, and secure.

Metadata Provenance. The history, origin, or transformation of data.

Metaheuristic Optimization. Solving complex search and optimization problems to find optimal or near-optimal solutions.

Data Preprocessing. The process of converting raw data into a clean dataset.

Bias Auditing. The formal process of inspecting and verifying a system to ensure that data are handled correctly and fairly.

Bias. A prejudice in favor of or against one thing, person, or group that causes machine learning models to produce unfair, inaccurate, or prejudiced outcomes.

Training Data. Dataset used to teach artificial intelligence and machine learning models to identify patterns, make decisions, or predict outcomes.

Traceability.

Accuracy.

Scalability.

CHAPTER 2

REVIEW OF RELATED LITERATURE

This chapter presents a review of related literature and studies relevant to the present research. It discusses theories, concepts, and findings from previous studies that are closely related to the variables of the study and provides a foundation for the research.

Dataset Bias as the Ethical Foundation of AI Systems

The widespread deployment of machine learning systems across critical sectors, including healthcare, finance, education, and criminal justice, has transformed dataset quality from a purely technical concern into a matter of ethical and social consequence. Biased training data can harm entire demographic groups even when models appear statistically accurate. This position is explicitly articulated in *Bias in Artificial Intelligence: Smart Solutions for Detection, Mitigation, and Ethical Strategies in Real-world Applications* by Samala and Rawas (2025), who define bias as a structural phenomenon embedded within datasets, algorithms, and social systems.

Types of Bias in AI Systems

Samala and Rawas (2025) identified three dominant forms of bias: gender, racial, and socio-economic, which originate from biased training data, algorithmic design choices, and historically rooted inequalities. Their work emphasizes that bias is not introduced at the modeling stage alone but enters the AI pipeline during data collection, curation, and labeling.

This structural framing is supported by *A Survey on Bias and Fairness in Machine Learning* by Mehrabi et al. (2021), which introduces a formal taxonomy of

dataset bias. The authors classify bias into representation bias, measurement bias, and historical bias. Representation bias arises when certain populations are underrepresented, measurement bias occurs when data collection tools perform inconsistently across groups, and historical bias reflects the continuation of social inequalities preserved in legacy datasets. These forms of bias often interact, making post hoc correction difficult.

Further reinforcing the socio-technical origins of dataset bias, Suddala (2021), in *Exploring the Ethical Implications of Biased Datasets on Decision-Making*, explains that bias emerges through historical inequities, sampling bias, measurement bias, and label bias. Label bias, in particular, arises from human subjectivity during annotation, embedding ethical distortions directly into ground-truth labels.

Most Used Attributes in Demographic Datasets

Demographic attributes are fundamental in detecting and evaluating bias in machine learning systems as they allow researchers to compare model performance across different groups. Based on the findings of multiple studies, gender, age, and race/ethnicity appear across all studies regardless of study domain or focus. These attributes function as the standard demographic attribute in fairness and bias research due to its availability and ease of annotation (Andrus et al. (2022); Cascone et al. (2026); Colacci et al. (2025); Dominguez-Catena et al. (2025); Majumdar et al. (2025); Sufian et al. (2025)).

Beyond the three, several studies also use socioeconomic status, education level, and nationality in demographic. (Colacci et al. (2025); Dominguez-Catena et al.

(2025); Majumdar et al.)). However, they are not common across all domains of machine learning and are only present when needed by the research focus.

Technical and Ethical Impacts of Biased Training Data

Dataset bias produces two distinct but interrelated harms: technical degradation and ethical injustice. Technical bias refers to failures in generalization, while ethical bias refers to systematically unfair outcomes for specific groups, even when overall accuracy remains high.

The technical consequences of biased datasets are empirically demonstrated in *Tackling Dataset Bias With an Automated Collection of Real-World Samples* by Sevetlidis et al. (2022). Results show that models trained on single-source datasets exhibited accuracy drops from approximately 70% to 48% when evaluated on external datasets. In contrast, models trained on multi-source datasets maintained stable performance across environments. This finding confirms that diversity in data sourcing is not merely ethically desirable but functionally necessary.

The ethical implications of biased datasets are emphasized in *Human-Centered Design to Address Biases in Artificial Intelligence* by Chen et al. (2023). The authors demonstrate that conventional evaluation practices often prioritize aggregate performance metrics, which conceal subgroup disparities. As a result, marginalized populations are disproportionately harmed even when systems appear statistically reliable.

These harms extend into organizational and industrial contexts. In *Biases in AI-Supported Industry 4.0 Research: A Systematic Review, Taxonomy, and Mitigation*

Strategies, Arévalo-Royo et al. (2025) show that dataset bias undermines not only predictive outcomes but also stakeholder trust, decision-making quality, and system legitimacy. Their phase-oriented framework reveals how bias propagates across the AI lifecycle.

Metadata Provenance and Lineage as Infrastructure for Ethical AI

While biased outcomes manifest at the model level, their origins lie in the hidden decisions embedded within data pipelines. Identifying these origins requires more than inspecting samples, it requires structured documentation of data origins, transformations, and contextual decisions. This is the role of metadata provenance and lineage.

In *The Impact of Modern AI in Metadata Management*, Yang et al. (2025) argue that metadata management has shifted from manual, rule-based systems to AI-driven automation. Traditional approaches, they note, are labor-intensive, inconsistent, and incapable of scaling with data complexity. By integrating NLP and machine learning, metadata processes such as data discovery, cataloging, and governance can be automated, enhancing transparency and accountability.

Metadata as an Ethical Instrument

The ethical role of metadata is empirically demonstrated by Boyd (2021) in *Exploring the Ethical Implications of Biased Datasets on Decision-Making*. When engineers were provided with detailed metadata, including sourcing criteria, demographic composition, and preprocessing steps, they identified ethical risks earlier and more reliably. This confirms that metadata is not merely descriptive, it is diagnostic.

Di Benedetto (2024), in *Using Metadata: The Ethical Problems*, formalizes this ethical function by defining five quality dimensions: accuracy, accessibility, coherence, completeness, and timeliness. He further argues that usability must be ensured at three levels: writer, reader, and repository, so that ethical assessments can be operationalized rather than merely documented.

Provenance for Explainability and Trust

In *Machine Learning Model Explainability Supported by Data Explainability*, de Oliveira et al. (2024) argue that model-level explainability is insufficient without data-level transparency. By linking model outputs to preprocessing transformations, provenance enables features to be traced back to raw sources. Without this linkage, explainability tools become opaque abstractions.

This governance role of provenance is further developed in *Data Lineage and Impact Analysis* by Karanam (2022), which positions lineage as the only reliable mechanism for root cause analysis in complex data ecosystems. Lineage enables proactive compliance, impact assessment, and change management.

Provenance in Regulated and High-Stakes Domains

Ahmed et al. (2023), in *Data Provenance in Healthcare: Approaches, Challenges, and Future Directions*, show that provenance bridges regulatory requirements with technical enforcement. However, they note that most existing approaches remain conceptual and fail to classify provenance records based on sensitivity.

This regulatory framing is echoed in *Building Ethical and Trustworthy AI Systems through Effective Metadata Management* by Sehgal and Bafna (2025), which

emphasizes that emerging regulations such as the EU AI Act require detailed documentation of system lifecycles, dataset sources, and preprocessing steps.

Scalability and Automation of Lineage

In *Hofmann's Tracer: A Machine Learning Approach to Data Lineage* (2020), automated lineage reconstruction is proposed for environments where manual documentation is infeasible. While effective for primary key detection, the system struggles with complex relational structures.

Paul (2024–2025), in *Data Lineage and Provenance Tracking for Multimodal ERP Analytics at Scale*, argues that traditional tools fail in heterogeneous environments and proposes unified metadata schemas and knowledge graphs.

The systemic consequences of undocumented lineage are exposed in *The Data Provenance Initiative* by Longpre et al. (2023), which identifies widespread license ambiguity, attribution errors, and geographic bias across 1,800 datasets.

Challenges in Bias Detection

Bias detection constitutes one of the most critical yet unresolved challenges in the development of ethical artificial intelligence systems. While prior themes establish that bias is structurally embedded within datasets and that metadata provenance enables its traceability, this theme focuses exclusively on the difficulties of identifying bias reliably, consistently, and at scale once it is already embedded in AI systems. The literature demonstrates that existing detection approaches remain methodologically unstable, computationally demanding, and weakly integrated into governance infrastructures.

Bias detection methods are designed to act as diagnostic instruments that reveal disparities in representation, association, and performance across demographic groups. These methods range from static word-embedding association tests to contextualized and dynamic evaluation frameworks. However, despite their widespread use, these tools often fail to produce consistent and reproducible results, undermining their practical usefulness.

In *Mind Your Bias: A Critical Review of Bias Detection Methods for Contextual Language Models*, Husse and Spitz (2022) conducted a systematic evaluation of widely used bias detection tools such as SEAT, CEAT, and LPBS. Their study revealed that bias measurements are highly sensitive to experimental configurations. Minor methodological changes, such as switching from personal names to group identifiers, modifying contextual prompts, or altering output encodings, can drastically alter bias scores for the same model. In several cases, these changes shifted results from statistically negligible to strongly significant.

Husse and Spitz (2022) further identify inconsistencies in code implementations and reporting practices across different studies. These inconsistencies result in reproducibility failures, where bias scores often reflect tool configuration rather than genuine model behavior. Consequently, current detection pipelines lack stability and standardization, making them unreliable for real-world deployment. This instability prevents bias detection from functioning as a dependable ethical safeguard.

Beyond technical fragility, governance-related limitations further complicate bias detection. In *Mitigating Bias and Advocating for Data Sovereignty: The Role of Metadata and Paradata in Ethical AI-Driven Information Systems*, Carrasco (2025) argues that detection mechanisms cannot function meaningfully in isolation from documentation, transparency, and accountability frameworks. She distinguishes between metadata, which documents dataset

attributes, and paradata, which records the procedural and decision-making processes that shape datasets. Without this dual-layer documentation, detection becomes fragmented, unverifiable, and difficult to enforce.

Carrasco (2025) further emphasizes that bias detection must align with sovereignty-oriented and regulatory principles, such as CARE (Collective Benefit, Authority to Control, Responsibility, and Ethics). This framing establishes that detection is not merely a technical operation but a governance obligation. When detection mechanisms are disconnected from policy frameworks and documentation systems, fairness assessments become symbolic rather than actionable.

The literature demonstrates that current bias detection approaches are methodologically unstable, computationally inefficient, and weakly embedded within governance infrastructures. These limitations prevent detection systems from operating as reliable ethical safeguards. As a result, bias detection requires scalable, standardized, and infrastructure-level solutions capable of functioning consistently in complex, real-world AI environments.

Bias Detection Tools Gap

Several studies have proposed and introduced different frameworks, algorithms, and methods in detecting biases in datasets whereas each has its own gaps and limitations that needs to be addressed and implemented.

Oveh and Isitor (2025) introduced a Bias Audit Framework (BAF) that detects and fixes algorithmic bias during the early stages of AI development. The framework uses four components: data bias assessment, model bias evaluation, developer training, and continuous monitoring. Using a healthcare dataset, applying BAF improved the

fairness while making sure that the model is just as accurate. The study highlights that fairness can be improved without sacrificing accuracy. However, the framework is not yet fully practical for real-world use as it faces a fairness vs accuracy trade-off and needs continuous monitoring. Furthermore, bias can still exist and reappear at different stages of the pipeline.

Li et al. (2025) proposed a method called “Decoupled Confident Learning” (DeCoLe) to identify mislabeled data in datasets while considering bias across different groups. Unlike traditional mislabeling detection, DeCoLe works on the assumption that labeling mistakes depend on both class and group. Results show that DeCoLe performs better than existing methods and that the improvement is especially strong when there is bias across groups. However, the method is limited to datasets that have group labels, struggles with underrepresented groups, and is harder to scale.

Bao and Barzilay (2022) proposed the “Learning to Split (LS)” algorithm that detects hidden bias by repeatedly splitting the dataset into train and test sets, training a model, and adjusting the split. Results show that it successfully found bias and challenging splits across different datasets. However, the LS algorithm exhibits notable limitations, such as the inability to explain what the bias is, is heavily model-dependent, and is computationally expensive.

Metaheuristic Optimization for Scalable Ethical Bias Detection

The detection of ethical bias within large-scale training datasets constitutes a computationally complex problem due to the high dimensionality, heterogeneity, and volume of metadata provenance. Exhaustive search methods become infeasible under these conditions, as the search space grows exponentially with the number of dataset attributes, transformations,

and lineage paths. As a result, bias detection in this context can be characterized as an NP-hard optimization problem. To address this, metaheuristic algorithms have been increasingly explored as scalable alternatives, as they balance global exploration and local exploitation to approximate optimal solutions within reasonable time constraints.

Metaheuristics operate by iteratively refining candidate solutions through stochastic search mechanisms. Global exploration allows the algorithm to traverse diverse regions of the search space, preventing early stagnation, while local exploitation refines promising solutions toward convergence. However, no single metaheuristic has been shown to consistently achieve an ideal balance between these two objectives across complex, high-dimensional problems.

This limitation is illustrated in *Energy-Aware Bag-of-Tasks Scheduling in the Cloud Computing System Using Hybrid Oppositional Differential Evolution-Enabled Whale Optimization Algorithm* by Chhabra et al. (2022). Their study demonstrates that standalone metaheuristics exhibit inherent performance trade-offs. The Whale Optimization Algorithm (WOA) is effective in local exploitation due to its bubble-net hunting strategy, which enables rapid convergence around promising solutions. However, this same behavior makes it prone to premature convergence in complex optimization landscapes. In contrast, Differential Evolution (DE) provides stronger global exploration through mutation and crossover mechanisms but often requires higher computational cost and exhibits slower convergence.

Chhabra et al. (2022) argue that this trade-off creates a performance threshold where no standalone algorithm consistently achieves both high solution diversity and fast convergence. As problem dimensionality increases, WOA tends to stagnate in local optima, while DE becomes computationally expensive for large-scale tasks. Although hybridization has been proposed to overcome these weaknesses, they note that many existing hybrid models fail to

provide a systematic improvement across convergence speed, diversity, and efficiency simultaneously.

The structural limitations of standalone swarm-based optimization are further examined in *Multistrategy Improved Whale Optimization Algorithm and Its Application* by Liu and Zhang (2022). Their analysis confirms that WOA suffers from “insufficient global search ability and slow convergence speed” when applied to complex optimization problems. While WOA is conceptually simple and computationally lightweight, it cannot independently resolve the imbalance between exploration and exploitation, especially in rugged and high-dimensional landscapes.

These observations are consistent with findings in *Differential Evolution: A Review of More Than Two Decades of Research* by Bilal et al. (2020). The authors emphasize that as optimization problems grow in dimensionality and complexity, it becomes increasingly difficult to identify the nature of the problem. This uncertainty makes standalone metaheuristics vulnerable to selecting inappropriate search strategies, often resulting in stagnation or inefficient convergence. Their review highlights that no single strategy universally dominates across problem classes.

Bilal et al. (2020) also observe that certain optimized standalone variants of DE, such as JADE and SaDE, have achieved competitive performance in international benchmarks, including IEEE CEC competitions. These variants rely on self-adaptive parameter tuning rather than full hybridization. They report that approximately 24% of documented performance improvements stem from modified mutation schemes alone. This suggests that, in some contexts, simpler adaptive strategies may rival complex hybrid models in terms of efficiency.

However, the problem of ethical bias detection in high-dimensional metadata provenance introduces unique challenges that may exceed the capabilities of standalone approaches. The search space in this context is not only large but also structurally complex, as bias can be embedded across multiple stages of a dataset's lifecycle. Random initialization strategies often fail to ensure sufficient coverage of this space, leading to incomplete or skewed detection results.

This motivates the need for enhanced optimization architectures that can traverse the entire data lineage while maintaining convergence efficiency. The performance limitations identified by Chhabra et al. (2022), Liu and Zhang (2022), and Bilal et al. (2020) collectively demonstrate that while standard metaheuristics provide a useful framework, they remain insufficient for large-scale, real-time ethical auditing tasks. These shortcomings form the basis for exploring hybridized approaches, which are examined in the succeeding theme.

Whale Optimization Algorithm (WOA) and Differential Evolution (DE) Hybridization

Hybridization has emerged as a dominant strategy in metaheuristic optimization to overcome the inherent limitations of standalone algorithms. Rather than relying on a single search behavior, hybrid models combine complementary strengths from multiple algorithms to enhance convergence speed, solution diversity, and robustness. In the context of large-scale and high-dimensional problems, such as ethical bias detection in metadata provenance, hybridization is particularly valuable because it enables both wide search coverage and precise local refinement.

The limitations of standalone metaheuristics motivate this shift. As discussed in *Energy-Aware Bag-of-Tasks Scheduling in the Cloud Computing System Using Hybrid Oppositional Differential Evolution-Enabled Whale Optimization Algorithm* by Chhabra et al.

(2022), WOA is prone to premature convergence due to its strong exploitation behavior, while DE, although effective in global exploration, incurs higher computational costs and slower convergence. These complementary weaknesses suggest that neither algorithm alone can consistently achieve optimal performance in complex optimization landscapes.

A concrete demonstration of hybridization is presented in *Multistrategy Improved Whale Optimization Algorithm and Its Application* by Liu and Zhang (2022). They proposed a hybrid model called the Differential Evolution Chaotic Whale Optimization Algorithm (DECWOA). This approach integrates DE operators into the WOA framework to improve global search ability while maintaining efficient local exploitation. Their method introduces “Sine chaos theory” to generate the initial population, increasing diversity and preventing early stagnation. Additionally, adaptive inertia weights are incorporated to dynamically balance exploration and exploitation throughout the optimization process.

Liu and Zhang (2022) reported that this hybrid structure allows the algorithm to avoid local optima more effectively than standalone variants. In complex engineering optimization tasks, the hybrid model achieved significantly higher accuracy, often approaching near 100%, where standard WOA and DE struggled. These results demonstrate that hybridization can systematically address the structural weaknesses of individual algorithms rather than merely tuning their parameters.

This observation aligns with the broader trends identified in *Differential Evolution: A Review of More Than Two Decades of Research* by Bilal et al. (2020). Their review highlights that hybridization has become increasingly prevalent over the past two decades, particularly in complex and high-dimensional problem domains. While they acknowledge that some adaptive standalone variants of DE (such as JADE and SaDE) perform competitively, they also

emphasize that hybrid models offer greater robustness when the nature of the optimization landscape is unknown or highly irregular.

In the context of ethical bias detection in metadata provenance, hybridization offers distinct advantages. Standard random initialization strategies often fail to distribute search agents uniformly across large and irregular search spaces. By incorporating chaotic mappings, such as the Sine chaos method discussed by Liu and Zhang (2022), a hybrid WOA–DE framework can ensure ergodic traversal of the solution space. This enables the algorithm to explore the full lifecycle of a dataset, increasing the likelihood of identifying hidden or subtle “audit gaps” where bias may be embedded.

Furthermore, the efficiency gains reported by Chhabra et al. (2022) suggest that hybridization can significantly reduce computational overhead. This is critical for making real-time ethical auditing feasible for massive training datasets. Rather than trading off between speed and accuracy, hybrid models aim to achieve both by design.

Bias Propagation in Data Preprocessing Pipelines

The data preprocessing phase is a critical point of failure where ethical distortions are often introduced before an algorithm even begins training. According to Victor (2022), the specific selection of preprocessing techniques, such as how categorical variables are encoded or how features are scaled, can directly determine whether a model output is fair or biased. This confirms that simply auditing the final results of a system is insufficient because the intermediate transformations themselves dictate the ethical outcome. This problem is often compounded by cascading bias, a phenomenon discussed by Hancox-Li (2020) where small, seemingly insignificant errors in early data cleaning stages grow into major discriminatory patterns by the time the data reaches the output stage. These findings establish that without a structured way to

trace the history or lineage of these edits, identifying the root cause of an unfair result is nearly impossible.

Black et al. (2023) argued that machine learning fairness is currently too theoretical and requires real-world practice across the entire ML pipeline. It was highlighted that current or traditional methods, such as adjusting models are no longer sufficient. They also emphasized that fairness problems must be addressed end-to-end, from problem definition to evaluation, in order to address the root cause of bias and fix it there.

Bias in Data Pipelines

A study by Ho & Beyan (2020) explained that bias can enter at every stage of the data pipeline. They emphasized that bias in data science is not just a model problem, but a pipeline problem. Six key stages were highlighted where bias can form which includes: Problem definition, where bias may form based on how the problem is posed. Data collection, which may exclude certain groups. Data Preparation, which may include subjective decisions on how to handle the cleaning, labeling, and preprocessing procedures. Modeling, where models learn patterns that reflect social biases. Evaluation, in which test data share the same biases as training data. Lastly, deployment and feedback, in which feedback from users may worsen the bias from an existing system over time. The paper suggests that being transparent about decisions and transformations, using diverse and representative datasets, evaluating models with fairness-aware metrics, and monitoring the system after deployment is the key to avoid biases.

Human Factor

Beyond the technical operations, the human factor remains a significant and often undocumented source of bias. Subjective decisions regarding what counts as an outlier or how to handle missing values can embed human prejudice directly into the cleaning scripts. Furthermore, the technical standards used in feature engineering can unintentionally skew results for specific demographic groups, such as age or race, if the transformations are not monitored for ethical parity (Koukaras & Tjortjis, 2025). Because traditional software debugging tools are designed to find functional code errors rather than ethical ones, there is a clear need for fairness debugging frameworks like Gopher to specifically target these preprocessing stages (ProQuest, 2023). These studies collectively justify the need for an automated system that can monitor both the technical and human-driven changes within a pipeline in real-time.

Synthesis

Dataset bias is not incidental. It is structural, persistent, and socially consequential. It can enter the data pipeline in the earlier stage, such as data preprocessing, making it not only a post-deployment error (Samala & Rawas (2025); Ho & Beyan, (2025)). Studies demonstrate that subjective human decisions and automated transformation silently introduce different kinds of bias in the early stage (Mehrabi, (2021); Suddala, (2021); Koukaras & Tjortjis (2025); ProQuest (2023)).

In bias detection, it's identified that age, race, and gender are the most used attributes in demographic dataset and usually used to determine bias. (Andrus et al. (2022); Cascone et al. (2026); Colacci et al. (2025); Dominguez-Catena et al. (2025); Majumdar et al. (2025); Sufian et al. (2025)). These biases propagate downstream, making them difficult to isolate, which is why

addressing the root cause of bias requires early-stage intervention rather than post deployment correction (Victor (2022); Hancox-Li (2020); Black et al. (2023)).

While existing bias detection frameworks attempt to address bias, they are met with limitations that hinder their performance. This includes evaluating only in later stages of the data pipeline, inconsistencies in implementation, methodologically unstable, bias reappearing at different stages of the pipeline, struggles with underrepresented groups, computationally expensive, leaves intersectional biases undetected, and lacks built-in mechanism for real-time correction (Husse & Spitz (2022); Carrasco (2025); Oveh & Isitor (2025); Li et al. (2025); Bao & Barzilay (2022)).

To achieve traceability and root-cause analysis in biased datasets, metadata provenance is advocated as an essential governance infrastructure to enhance transparency and accountability (Yang et al. (2025); Boyd (2021); de Oliveira et al. (2024); Karanam (2022)). However, this generates massive, high-dimensional metadata logs that need to be searched for hidden biases.

Metaheuristic algorithm, like the hybridization of Whale Optimization Algorithm and Differential Evolution (WOA-DE) offer a scalable solution to navigate this complexity as they balance global exploration and local exploitation to approximate optimal solutions (Chhabra et al. (2022); Liu & Zhang (2022); Bilal (2020)).

To address the gaps mentioned, the study introduces a bias auditing tool that monitors the preprocessing stage of the data pipeline using metadata provenance. With the use of the WOA-DE algorithm, the cause of bias can be pinpointed by going through the metadata recorded. This approach offers an opportunity to elevate the efficiency of a scalable, real-time auditing system within data pipelines.

CHAPTER 3

METHODOLOGY

This chapter contains the research methodology employed, the sources of data, instruments used, system architecture, data generation procedure, ethical considerations, including the data analysis and statistical treatment used in the study.

Research Design

The researchers will employ an experimental research design to achieve the thesis' aims, specifically focusing on the development and evaluation of the proposed bias audit system. According to Sreekumar (2024), experimental research is used to determine cause-and-effect relationship between variables and to test hypotheses. This approach is well-suited for this study because it involves measuring technical performance through data and testing the effectiveness of the hybrid WOA-DE algorithm. Since the main goal is to solve a computing problem related to ethics and data pipelines, the research is structured into several technical stages that contribute to the final solution.

The study also follows the Input-Process-Output (IPO) model to ensure that every stage of the study is organized and measurable. In the initial phase, the researchers will focus on the preparation of structured datasets and the assignment of protected demographic attributes. This is followed by the development and implementation of the active provenance system and the hybrid algorithm which serves as the core technical process. Finally, the system will be evaluated based on its ability to generate bias flags, provide mitigation suggestions, and maintain processing speed compared to traditional methods.

Through this structured approach, the researchers can effectively answer the statement of the problem and verify the hypotheses regarding system accuracy and performance. By recording and analyzing the performance data generated during these stages, the study will provide a clear and evidence-based conclusion on the effectiveness of using metadata for bias detection.

Sources of Data

The data for this study will be sourced from a publicly available dataset obtained from the UC Irvine (UCI) Machine Learning Repository. The dataset utilized is the Adult Income Dataset, also referred to as the "Census Income" dataset, originally extracted from the 1994 U.S. Census Bureau database by Ronny Kohavi and Barry Becker. It contains a combined 48,842 instances from the train and test data and 14 attributes, with a binary target variable indicating whether an individual's annual income exceeds \$50,000.

Sampling Data

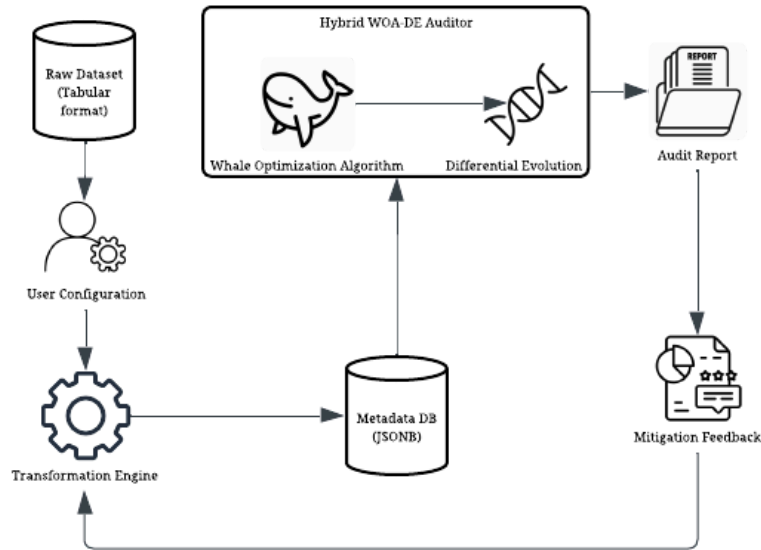
The sampling method used in this study is purposive sampling. It allows for the selection of datasets that strictly align with the research objectives. The datasets included in the study must meet the following criteria:

1. The dataset must have at least three protected attributes in variables or columns.
2. The dataset must contain a binary target variable to allow the calculation of success rates for the fairness metrics.
3. The dataset must be publicly available or distributed under terms that permit legal use, reproduction, and analysis for research purposes.
4. The dataset must be in the tabular format.

System Architecture

The proposed architecture represents an integrated, closed-loop framework for automating the lifecycle of bias detection and mitigation within data pipelines.

Figure 3.1: System Architecture



The system architecture works as follows. Before preprocessing begins, the framework accepts a User Configuration, allowing auditors to initialize the target variable and the specific protected attributes they intend to monitor for disparities. First, the raw datasets are processed by the transformation engine, cleaning the data and ensuring consistent format across all records. Any modifications are recorded into the JSONDB database. Our system utilises a semi-structured JSONB format to ensure computational scalability and schema flexibility. This avoids the overhead of sparse relational tables and allows the Hybrid WOA-DE algorithm to perform high-speed, keyed lookups on diverse transformation attributes. Then, the cleaned dataset goes to the hybrid WOA-DE auditor, which works as shown in Figure 3.2 and 3.3. The

findings are compiled in the audit report, which subsequently generates Mitigation Feedback. This recursive feedback loop enables the automated or guided adjustment of preprocessing parameters, providing a pathway for the automatic correction of identified biases in future runs.

Figure 3.2: Whale Optimization Algorithm

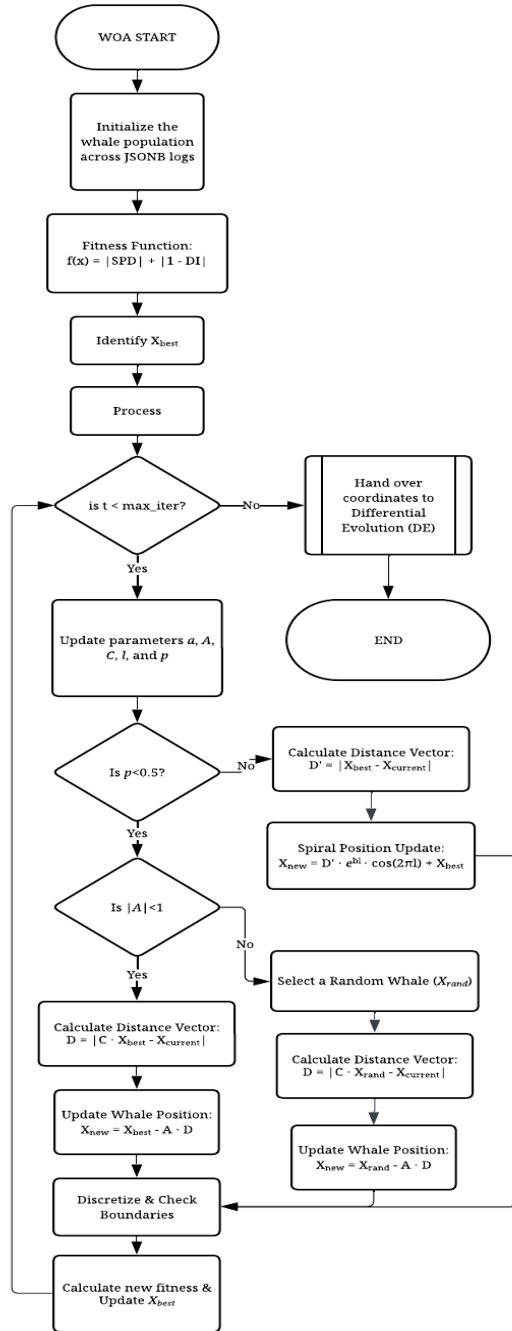
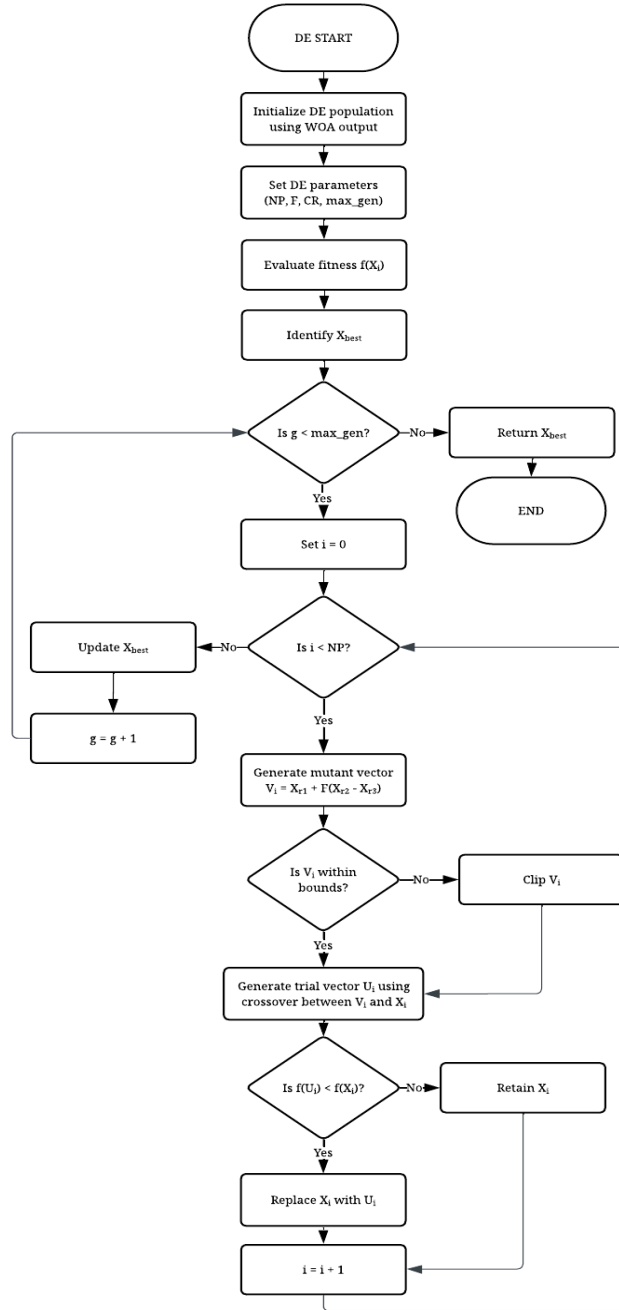


Figure 3.3: Differential Evolution



The flowchart above depicts the process of hybridization of the WOA-DE algorithm. The process will always start at WOA, the algorithm starts by initializing the population of whales scattered across the JSONB database, with their position vectors representing specific queries to different historical data transformation logs and demographic attributes. Next, the algorithm calculates the fitness of each search agent. This process involves retrieving the pre-calculated summary statistics from the metadata log and running the bias formula. The whale that discovers the highest bias score is designated as the current best search agent. After that, the algorithm enters a loop that runs until the maximum number of iterations is reached. Inside this loop, the algorithm updates its parameters for every single search agent. The parameter a acts as a countdown, decreasing linearly from 2 to 0 to tighten the search over time, while p is a random probability that decides the whale's next movement into three distinct paths based on the values of p and the tuning coefficient A .

- **Spiral Bubble-Net Attack (Exploitation):** If $p \geq 0.5$, the algorithm moves to the spiral updating position. The whale use logarithmic spiral equation to search the immediate “neighborhood” wrapped closely around high bias metadata log found by X^* .
- **Encircling Prey (Exploitation):** If $p < 0.5$ and the absolute value of $A < 1$, the whales use the shrinking encircling mechanism. This forces the search agents to take small, precise steps to tighten their search net closer to the known ethical violation in the pipeline.
- **Search for Prey (Exploration):** If $p < 0.5$ and $A \geq 1$, the algorithm triggers the exploration phase. Instead of moving toward the known bias, the whales update their positions based on a completely random chosen whale from the population. This crucial step makes our algorithm search globally across the entire JSONB database so it does not get trapped in a local optimum and miss other unfair pipeline steps.

After the movement phase, the algorithm regroups to check if any whale has gone beyond the search space boundaries and amends it. The fitness function is recalculated, and if a higher bias is found, the X^* is updated. Once the maximum iterations are completed, the WOA terminates and outputs the coordinates of the highest-bias metadata clusters and hands them over to the Differential Evolution.

The Differential Evolution algorithm receives the coordinates from the Whale Optimization Algorithm and uses them to initialize its population. This ensures that DE does not start from scratch but instead begins its search from a promising region of the solution space already identified by WOA. The algorithm then sets its necessary control parameters: NP, which is the population size, F, which is the scaling factor, CR, which is the crossover rate, and max_gen, which is the maximum number of generations. After that, the fitness of each individual in the population is evaluated, and the best solution is identified.

The algorithm then enters its outer loop, which runs until the maximum number of generations is reached. If the condition is met, the algorithm returns the best solution and terminates. Otherwise, it cycles through the entire population, processing each individual sequentially within the inner loop.

For each individual, DE performs three sequential operations:

- **Mutation:** A mutant vector is generated by selecting three distinct individuals at random from the current population. It calculates the difference between two of them, scales this difference using the scaling factor, and adds it to the third individual. This creates a new directional search point. If this new point falls outside the search space, it is clipped to the nearest boundary so it can remain a valid potential solution.

- **Crossover:** A trial vector is produced by combining the mutant vector and the original individual based on the crossover rate. Each coordinate of the trial vector is drawn from the mutant vector with a probability defined by the crossover rate, and the remaining coordinates are taken from the original individual, to ensure the offspring inherits traits from both vectors.
- **Selection:** The fitness of the trial vector is evaluated and compared against the current individual. If the trial vector yields a better fitness score, it replaces the current individual in the population. If not, the current individual is retained with no changes.

After all individuals in the population have been processed, the best solution is updated if a better one has been found, and the generation counter is incremented by 1. Then, the outer loop repeats until the maximum generation is reached. Afterwards, the algorithm returns the refined coordinates of the highest-bias metadata clusters as its final output.

Research Instrument

This study uses a system developed by the researchers as the primary instrument for data generation and collection. The system processes raw datasets through data pipelines and analyzes them for potential bias. Its performance will be recorded and analyzed to determine whether bias can still emerge after data has been cleaned. Through these generated performance data, the system supports both the collection of relevant data and the evaluation of the proposed solution. The following tools were used to develop the system:

1. **Python:** Serves as the primary programming language for developing the tracking decorators, simulating the data pipelines, and executing the Hybrid WOA-DE algorithm.
2. **PostgreSQL(JSONB):** Utilized as the database management system. The semi-structured JSONB format is specifically chosen to store the metadata provenance

logs, allowing for schema flexibility and high-speed data retrieval during the algorithmic auditing phase.

3. **Git:** Employed for version control to maintain the integrity of the system's development lifecycle.

Data Generation/Gathering Procedure

The research utilized a secondary data gathering procedure by acquiring an existing structured dataset from the UC Irvine Machine Learning Repository. The selection was based on a purposive sampling criteria, requiring the data to contain the most common demographic attribute: age, sex, and race.

Ethical Considerations

The researchers ensure that the following will be observed throughout the study:

1. Privacy and Confidentiality

All data and information collected and used in the development of the system will be handled responsibly and in accordance with the Data Privacy Act of 2012 (Republic Act No. 10173). Any personal information collected by the system will be confidential.

2. Responsible Use of Data

All data used in the development, testing, and evaluation of the bias detection system will be strictly for academic and research purposes only. Sensitive information will only be accessible to authorized users, including the researchers.

3. Proper Acknowledgment of Sources

All intellectual property and sources of information, related literature and studies, software tools, and programming used in system development will be acknowledged through proper citation and referencing to avoid plagiarism.

4. Ethical Use of the System

The system developed in this study is strictly for its stated purpose only. It will not be designed or used in ways that it could cause harm, violate privacy, misuse information, or compromise security.

Data Analysis (Procedure and Treatment)

To evaluate the effectiveness of the proposed framework, the analysis is divided into three distinct phases aligned with the research objectives:

Phase 1: Traceability and Provenance Accuracy

To evaluate the system's traceability, the researchers will conduct a *Lineage Reconstruction Test*.

- **Procedure:** Known biases will be intentionally "injected" during specific pipeline transformations (e.g., during null-value imputation).

The analysis will measure the Detection Accuracy Rate by comparing the system's "Audit Flags" against the Ground Truth (the known injected bias). This will verify if the system can successfully identify the exact transformation responsible for the ethical shift.

Phase 2: Algorithmic Performance Evaluation

The efficiency of the Hybrid WOA-DE algorithm will be analyzed using a two-stage evaluation:

- **WOA Scouting Efficiency:** We will measure the Convergence Speed of the Whale Optimization Algorithm as it scouts for suspicious neighborhoods within the high-dimensional metadata.
- **DE Pinpointing Accuracy:** The researchers will analyze the Root Mean Square Error (RMSE) or Precision-Recall metrics of the Differential Evolution phase to determine how accurately it isolates the specific record that caused the bias.

The data will be visualized using Convergence Graphs and Heatmaps to show the algorithm's path from global scouting to local pinpointing.

Phase 3: Computational Efficiency and Scalability Analysis

To assess the operational efficiency of the proposed framework, the researchers will conduct a comparative performance test.

- **Procedure:** Microsoft Fairlearn will serve as the baseline model to perform standard bias auditing on the raw, original dataset. This allows the researchers to observe the difference in processing time between a traditional raw-data scan and the proposed metadata-based approach using JSONB logs.
- **Scalability Stress Test:** Both the proposed system and the baseline will be tested across varying dataset volumes, ranging from 10,000 instances to the full volume of the UCI Adult Income dataset, to monitor how performance scales as the search space grows.

- **Evaluation Metrics:** The primary metric for this phase is Audit Latency, which measures the total execution time in seconds. The data will be analyzed to determine if there is a statistically significant difference in speed.

Statistical Treatment

The statistical methods that will be implemented in this study will be in the form of fitness function. In machine learning and optimization algorithms, a fitness function is a mathematical evaluation tool that measures exactly how "good," "fit," or successful a particular solution or state is. It is the specific mechanism used to guide the algorithm's search through a complex problem. The following methods will be implemented:

1. Statistical Parity

Statistical parity(also known as Demographic parity) is a definition of fairness in machine learning and data analysis where the probability of a positive outcome is equal across all protected demographic groups. It ensures the algorithm's decisions are independent of sensitive attributes, aiming to prevent discrimination by equating selection rates.

To satisfy statistical parity, the probability of a positive prediction must be equal across groups. Mathematically, this is expressed as:

$$P(\hat{Y} = 1|A = a) = P(\hat{Y} = 1|A = b)$$

Where:

- $\hat{Y} = 1$ represents the predicted positive/favorable outcome.

- A represents the sensitive demographic attribute (e.g., race or gender).
- a and b represent the distinct groups being compared

When actually auditing a dataset, algorithms compute the Statistical Parity Difference (SPD) to quantify exactly how large the bias gap is:

$$SPD = P(\hat{Y} = 1|A = u) - P(\hat{Y} = 1|A = p)$$

Where:

- $A = u$ represents the unprivileged (historically disadvantaged) group
- $A = p$ represents the privileged group

The standard formula calculates the difference between the favorable outcome rate of the unprivileged group and the privileged group.

2. Disparate Impact

Disparate impact(also known as adverse impact) occurs when a policy, practice, or algorithm that appears completely neutral on the surface disproportionately harms or excludes a protected demographic group.To quantify this bias within a dataset, auditors use the Disparate Impact metric, which determines the ratio of favorable outcomes that different groups encounter.

$$DI = \frac{P(\hat{Y} = 1|A = u)}{P(\hat{Y} = 1|A = p)}$$

Where:

- $P(Y=1 | A=u)$ is the probability of a positive prediction for the unprivileged group.
- $P(Y=1 | A=p)$ is the probability of a positive prediction for the privileged group.

To evaluate whether this mathematical ratio is problematic, the researchers will rely on the "four-fifths rule" (also called the 80% rule). It flags potential discrimination when a protected group's selection rate falls below 80% of the highest group's rate.

- A ratio of exactly 1.0 means perfect parity, where both groups receive positive outcomes at the exact same rate
- If the selection rate for a minority group falls below four-fifths of the rate of the majority group, the practice is generally flagged as having an adverse impact.

3. Fitness function

The fitness function will be the combined formula of Statistical Parity Difference and Disparate Impact:

$$f(X) = |StatisticalParityDifference| + |1 - DisparateImpact|$$

If the data is perfectly fair, SPD will equal to 0 and the DI will equal to 1. The equation becomes 0 otherwise if the data has bias the result will be > 0 .

Though Statistical Parity and Disparate Impact can be used on any categorical variable in a dataset (Bird et al., 2020), they are primarily defined and applied to

protected attributes like race, gender, or age to detect unfair discrimination in AI models or decision systems (La Malfa et al., 2024).

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